

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

During a portion of a flight, a small airplane's cruising speed varied between **150** miles per hour and **170** miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

Answer

- A. $s \leq 20$
- B. $s \leq 150$
- C. $s \leq 170$
- D. $150 \leq s \leq 170$

Correct Answer: D

Rationale

Choice D is correct. It's given that during a portion of a flight, a small airplane's cruising speed varied between **150** miles per hour and **170** miles per hour. It's also given that s represents the cruising speed, in miles per hour, during this portion of the flight. It follows that the airplane's cruising speed, in miles per hour, was at least **150**, which means $s \geq 150$, and was at most **170**, which means $s \leq 170$. Therefore, the inequality that best represents this situation is $150 \leq s \leq 170$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.